Date :
P.No. :

(4) $P(M) = 0.05$

here - M = Patient has Meningitis

$-M$ = Patient does not have meningitis

$+$ = Positive test

$-$ = Negative test

$P(M) = 0.05$

$P(-M) = 0.95$

- Test accuracy $P(+|M) = 0.95$
- Test gives negative for healthy people 70%

$$P(-|-M) = 0.70$$

$$P(+|-M) = 1 - 0.70 = 0.30$$

Probability of Meningitis given a positive test

$$P(M|+) = \frac{P(+|M)\,P(M)}{P(+)}$$

$$P(+) = P(+|M)\,P(M) + P(+|-M)\,P(-M)$$

$$P(+) = (0.95 \times 0.05) + (0.30 \times 0.95)$$

$$\Rightarrow 0.3325$$

$$P(M|+) = \frac{0.95 \times 0.05}{0.3325}$$

$$\Rightarrow 0.143 \Rightarrow 14.3\%$$

Probability patient actually does not have meningitis = 85.7%

Now update the probability ~

$P(M|+) = 0.143$

$P(-M|+) = 0.857$

$P(-|M) = 1 - 0.95 = 0.05$

$P(-|-M) = 0.70$

Date :
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$$P(-) = P(-|M)\,P(M|+) + P(-|-M)\,P(-M|+)$$

$$\Rightarrow (0.05 \times 0.143) + (0.70 \times 0.857)$$

$$P(-) = 0.607$$

$$P(M|+,-) = \frac{P(-|M)\,P(M|+)}{P(-)}$$

$$\Rightarrow \frac{0.05 \times 0.143}{0.607} \Rightarrow 0.0118$$ Ans